

CURRICULUM VITAE

Waylon J. Howard, Ph.D.

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Professional Summary

Quantitative research and measurement scientist with 18 years of experience across 8 institutions studying how complex human experiences can be measured, modeled, and understood over time. Bridges advanced quantitative methodology with practical research strategy across clinical, behavioral, health, and translational science. Expertise spans research design, psychometrics and measurement science, longitudinal and multilevel modeling, structural equation modeling, causal inference, predictive modeling, and other advanced quantitative methods. Partners with scientific leaders and multidisciplinary teams to shape research questions, develop rigorous study designs, and translate complex data into meaningful evidence. Proven record of providing methodological and scientific leadership across NIH, PCORI, CDC, and industry-funded research programs collectively representing more than \$20M in awarded funding, while advancing research from conception through dissemination and mentoring the next generation of quantitative researchers.

Core Expertise

Research & Scientific Strategy

Research Strategy | Scientific Strategy | Research Design | Research Program Development | Evidence Generation | Grant & Funding Strategy | NIH/PCORI Research Strategy | Research Infrastructure

Measurement & Quantitative Science

Measurement Science | Psychometrics | Construct Validation | Factor Analysis & Structural Equation Modeling | Longitudinal & Multilevel Modeling | Causal Inference | Missing Data Methods | Predictive Modeling

Behavioral, Clinical & Applied Research

Behavioral Science | Mental & Behavioral Health Research | Clinical Research | Digital Health | Patient-Centered Outcomes | Translational Science | Observational & Experimental Research | Real-World Studies

Scientific Computing & Reproducible Research

R | Python | SAS | Mplus | SQL | Statistical Computing | Reproducible Research Workflows | Research Applications | Shiny

Scientific & Methodological Leadership

Methodological Leadership | Scientific Leadership | Cross-Functional Collaboration | Multidisciplinary Research Leadership | Investigator Mentorship | Team Development | Research Capacity Building

Selected Career Highlights

- **Led quantitative and methodological strategy across a \$20M+ portfolio** of NIH, PCORI, CDC, and foundation-supported clinical, behavioral, and translational research programs.
- **Provided scientific and methodological leadership across multidisciplinary research teams**, shaping research questions, study design, measurement strategies, advanced quantitative analyses, grant development, and dissemination.
- **Built research infrastructure supporting 8-26 grant submissions annually**, strengthening institutional capacity for rigorous research design, quantitative planning, methodological review, and reproducible science.
- **Authored 37 peer-reviewed publications with 1315 citations (h-index, 17; i-10 index, 19)** with 1300+ citations, contributing to research spanning pediatric health, behavioral and mental health, development, and translational science.
- **Developed reproducible research workflows and quantitative applications** that improved analytical rigor and efficiency while creating reusable methodological infrastructure across research programs.

Research Appointments

Principal Biostatistician

Seattle Children's Research Institute, Seattle, WA | 2022-Present

- **Lead quantitative research and methodological strategy** across clinical, behavioral, developmental, and translational science programs, shaping research questions, study design, measurement strategies, and approaches to evidence generation.
- **Provide senior scientific and methodological leadership** across NIH, PCORI, CDC, and foundation-funded research programs, contributing to **\$20M+ in awarded research funding and an additional \$14M in pending funding opportunities**.
- **Develop and apply advanced quantitative methods to complex questions about human health and behavior**, including longitudinal and multilevel modeling, structural equation and latent-variable methods, psychometrics and measurement science, finite mixture modeling, conditional process modeling, and new methodology development for longitudinal dyadic modeling.
- **Build reusable methodological infrastructure**, including analytic frameworks, reproducible workflows, and quantitative research applications that strengthen rigor and extend research capabilities across projects and programs.
- **Partner with clinicians, investigators, epidemiologists, data scientists, and scientific leaders** to translate complex research questions into rigorous study de-

signs, analytic strategies, scientific publications, and competitive funding proposals.

- **Mentor investigators and quantitative researchers** in research design, measurement, advanced quantitative methods, grant development, reproducible analysis, and scientific communication.
- **Authored 20+ peer-reviewed publications since 2024** contributing to collaborative scholarship spanning behavioral, developmental, clinical, and translational research.

Director of Research, Patient-Centered Outcomes

Open Health, Bethesda, MD | 2021-2022

- Directed global quantitative analytics operations supporting oncology real-world evidence programs and \$12M annual revenue.
- Led a 10-person US/India analytics organization delivering statistical programming, data analytics, and evidence generation solutions.
- Built analytics infrastructure supporting transition from traditional biostatistical workflows to scalable SQL/cloud-based delivery pipelines.
- Partnered with clinical, product, engineering, and commercial teams to operationalize analytics capabilities during rapid organizational growth.
- Developed automated R/Shiny tools reducing analyst training time by approximately 40%.
- Established standardized analytic processes, SOPs, and quality frameworks improving reproducibility and delivery consistency.

Director of Biostatistics and Data Analytics

ConcertAI, Boston, MA | 2019-2021

- Directed a 10-person global analytics team (US and India) supporting **\$12M in annual revenue** during rapid organizational growth and platform transition.
- **Scaled analytics operations with zero delivery disruption** during transition from primarily biostatistics via SAS to data management and analytics via SQL with AWS based delivery pipelines.
- Developed R/Shiny workflow tools, **reducing staff training time by ~40%**
- Co-developed 15+ SOPs to standardize analytics delivery and improve team performance
- Managed global project delivery, onboarding, and resource allocation across distributed teams

Director of Research in Biostatistics and Data Management Associate Professor

Atrium Health, Charlotte, NC | 2018-2019

- Directed biostatistics (6 staff) and data management (4 staff) teams supporting real-world evidence and patient-centered outcomes research.
- Expanded institutional research capacity through a 6-course training series reaching **150+ medical residents**
- Oversaw study design, analysis, and reporting across a diverse clinical portfolio
- **Increased accepted abstracts and publications by 37%** and presentations by 10%

Director of Biostatistics, Epidemiology, Econometrics, and Programming Core

Seattle Childrens Research Inst., Seattle, WA | 2015-2018

Led a 5-person biostatistics and analytics research support core with a **\$600K operating budget**, supporting institutional research and grant development.

- Developed automated workflow and capacity management tools (across R, Qualtrics, Asana) to improve efficiency and transparency.
- Expanded Core supported NIH grant submissions from 10 to **60+ annually**, contributing to **~\$5M in new funding**
- Strengthened core infrastructure to support growing institutional demand

Faculty Research Methodologist

Institute for Social Science Research

University of Massachusetts, Amherst, MA | 2014-2015

- Directed methodology training and consulting services supporting research across seven colleges.
- Led a team of 5 consultants delivering internal research support.
- Organized a 12-course advanced methodology conference training **124 participants**
- Expanded institutional capacity in research design and analytic methods

Director of Research and Evaluation

Maryland Ctr. for Developmental Disabilities

Kennedy Krieger Institute, Baltimore, MD | 2012-2014

- Led research, evaluation, and policy-focused initiatives to improve outcomes for individuals with developmental and intellectual disabilities.
- In collaboration with the Senior Director led community-based research, program evaluation, and needs assessments

- Supported multidisciplinary efforts spanning public health, policy, and service delivery

Visiting Assistant Professor

Johns Hopkins Bloomberg School of Public Health, Baltimore, MD | 2014

- Contributed to graduate teaching and research in public health methodology, disability research, and epidemiology.

Affiliated Faculty Member

Wendy Klag Center for Autism and Developmental Disabilities,

Johns Hopkins University, Baltimore, MD | 2014

- Collaborated on interdisciplinary research and training initiatives in autism and developmental disabilities.

Co-director of the Knowledge Acquisition and Scientific Inquiry Core

Kennedy Krieger Institute, Baltimore, MD | 2013-2014

- Led development of curriculum and training programs to strengthen research literacy and methodological rigor in special education and research professionals.

Senior Research Data Analyst

Biostatistics, Epidemiology, and Data Management Core,

Johns Hopkins University, Bayview, MD | 2013-2015

- Provided statistical design, data management, and analytic support across funded research programs.
- Contributed to **10+ publications, presentations, and invited talks**
- Served as methodological consultant on grant-funded program evaluations

Quantitative Psychology Research Assistant

Juniper Gardens Children's Project

University of Kansas, Kansas City, KS | 2008-2012

- Provided research methods and data analysis support for community-engaged research on child development, early literacy, and behavioral interventions.
- Contributed to the **development of the Early Communication Indicator (ECI)**, a standardized early childhood language development tool.

Quantitative Psychology Research Assistant

Ctr. for Research Methods and Data Analysis

University of Kansas, Lawrence, KS | 2010-2011

- Provided methodological consulting and analytic support across academic departments and external partners.
- Assisted with research design, statistical analysis, and software training workshops

Education

Ph.D. Quantitative Psychology, University of Kansas (2012); Advisor: **Todd D. Little**
Dissertation Using principal component Analysis to obtain auxiliary variables for missing data in large data sets.

- Developed scalable methods for handling missing data using PCA-based auxiliary variables

Selected coursework: Research Design, Factor Analysis, Nonparametric Methods, Test Theory, Multivariate Analysis, Multilevel Modeling, Categorical Data Analysis, Structural Equation Modeling, Longitudinal Structural Equation Modeling, Item Response Theory, Advanced Quantitative Methods, Simulation & Monte Carlo Studies, Dynamical Systems & Intraindividual Change

M.S. Experimental Psychology, Emporia State University (2007) *Thesis* Housing effects on ethanol consumption: free access and operant situations.

Designed and conducted animal experiments examining behavioral and environmental influences on substance use

Co-managed an animal research laboratory

B.A. Psychology, Lyon College (2005)

Completed international study and research experiences across Australia, Spain, Brazil, Venezuela, and Finland

Awards and Honors

Patterson Decade Award (2014) Lyon College

Dissertation Award (2012) Awarded by the Society of Multivariate Experimental Psychology

Best Student Paper (2006) Awarded by the Association for Psychological and Educational Research in Kansas for: Howard (2006). Effects of housing on consumption ratio and operant responding ratio in rats: Ethanol vs. water.

NSF Research Experiences for Undergraduates (REU) Fellow, Oklahoma State University (2003). Sponsor: Charles I. Abramson, PhD

Selected for competitive NSF-funded program; research resulted in two peer-reviewed publications

Professional Publications

In progress

1. Irvin, D. W., Sherlock, P., Hegde, V., & Howard, W. J. (in preparation). Modeling dynamic valued social networks with the temporal generalized exponential random graph model [Unpublished manuscript].
2. Kroshus-Havril, E., Kahn, N., Steiner, M. K., Christakis, D., & Howard, W. (in preparation). Development of brief scale of psychological need satisfaction with screen use for adolescents.
3. Bonilla-Velez, J., Kosai, H., Masco, V., & Howard, W. J. (in preparation). Structural validity and preliminary known-groups evidence for a caregiver perspectives survey in pediatric otolaryngology.
4. Clifton, C. R., & Howard, W. J. (in preparation). Examining the relationship between substance use and suicide among Black pediatric patients.
5. Taylor, M. R., **Howard, W. J.**, Barton, K. S., Bradford, M. C., Cardenas, E., Wittig, J., Karthik, S., Dhawan, S., Fladeboe, K. M., Baker, K. S., & Rosenberg, A. R. (in preparation) Psychosocial Phenotyping of Adolescents and Young Adults with Cancer Using Synchronous Qualitative, Quantitative, and Digital Biomarker Data.

Submitted

1. Ormanidou, I., **Howard, W. J.**, Bijlani, K., Denno, D., Amoako-Yirenkyi, P., Heike, C., Adusei, A., Oliphant, H., Crerand, C., Jackson, O., Discolo, C., Vetter, S., Wilson, K., Geronimo, Y., Galera, R., Lyman, N., Whyte, V., Pajor, N., & McKinney, C. Predicting underweight in infants with cleft lip and/or palate in 10 U.S. children's hospitals [Manuscript submitted for publication].
2. Kroshus, E., Garrett, K., Steiner, M. K., Matovina, S., McCleery, J., Turner, K., Senturia, K., **Howard, W. J.**, Kinard, D., Nieto, C., & Tandon, P. Building school capacity for physical activity: Evaluation of the Physical Activity Coordinator Program [Manuscript submitted for publication].
3. Kroshus-Havril, E., Garrett, K., Steiner, M. K., Matovina, S., McCleery, J., Turner, K., Senturia, K., **Howard, W. J.**, Kinard, D., Ellerton, L., Nieto, C., & Tandon, P. S. Formative and feasibility evaluation of the Physical Activity Coordinator Program [Manuscript submitted for publication].
4. Paek, B. W., Keller, S. B., **Howard, W. J.**, Dorn, M., & Walker, M. P. R. Introducing a calculator for prediction of donor demise on POD#1, and of donor live birth, after laser for TTTS [Manuscript submitted for publication].
5. Paek, B. W., Keller, S. B., **Howard, W. J.**, Dorn, M., & Walker, M. Clinical significance of new-onset abnormal ductus venosus waveform in donor fetuses after laser surgery for twin-to-twin transfusion syndrome: A cohort study [Manuscript submitted for publication].

6. Villegas, S., **Howard, W. J.**, Steinbrenner, J., Nowell, S., Irvin, D., Buzhardt, J., Boyd, B. Exploring variation in expressive communication between Latino and non-Latino infants and toddlers with autism. [Manuscript submitted for publication]
7. Olson, B., Buddhe, S., Tressel, W., Young, L., Deen, J., Vu, T, **Howard, W. J.**, Soriano, B. Factors Impacting Time Spent on a First-Time Echocardiogram: Is There Disparity Between Races and Ethnicities? [Manuscript submitted for publication]
8. Saelens B. E., Rowland, M. G., Ibrahim, M. A., **Howard, W. J.**, Seerha, H., Melhorn, S. J., Schur, E. A., Roth, C. L. Patterns of Children's Weight Change During Family-Based Weight Management Intervention and Associations with Outcome Through Follow-up. [Manuscript submitted for publication]
9. Brown, N. K., Romberg, E. K., Pinto, N. M., **Howard, W. J.**, Robinson, J. D., Soriano, B., & Buddhe, S. Establishing the reliability of non-contrast 4D flow in congenital heart disease [Manuscript submitted for publication].
10. Querdasi, F. R., Murray, C., Johnson, P., Murphy, T. B., Levy, R. L., van Tilburg, M., Kuklinski, M., Bailey, J., **Howard, W. J.**, Diakhate, H., Daniels, K., Badillo, I., & Palermo, T. M. Family risk and resilience patterns differentiate early childhood vulnerability to intergenerational transmission of abdominal pain [Manuscript submitted for publication].
11. Pierce, M., **Howard, W. J.**, Niu, I., Kim, S-J., Salehi, P., Relationship between Aberrant Behavior Checklist (ABC) scores, hyperphagia, and quality of life in youth with Prader-Willi syndrome. [Manuscript submitted for publication]
12. Paek, B. W., Keller, S. B., **Howard, W. J.**, Dorn, M., & Walker, M. Clinical significance of new-onset abnormal ductus venosus waveform in donor fetuses after laser surgery for twin-to-twin transfusion syndrome: A cohort study [Manuscript submitted for publication].
13. Paek, B. W., Keller, S. B., **Howard, W. J.**, Dorn, M., & Walker, M. P. R. Introducing a calculator for prediction of donor demise on POD#1, and of donor live birth, after laser for TTTS [Manuscript submitted for publication].

Completed

1. Murray, C., Li, R., Stone, A., **Howard, W. J.**, & Palermo, T. (2026). Change in pain phenotypes from adolescence to young adulthood: A latent class and transition analysis. *The Journal of Pain*, 41. [Impact Factor 4.4], Cite, Source Document)
2. Little, L. M., Keith, S. N., Wallisch, A., Rous, B. S., Johnson, T., **Howard, W. J.**, & Tomchek, S. (2026). Age of entry into early intervention: Influence of the COVID-19 pandemic, telehealth, and other factors. *The Journal of Pediatrics: Clinical Practice*, 200207. [Impact Factor 1.0] (PDF, Cite, Source Document)
3. Ittleman, B., **Howard, W. J.**, Virk, K., Brown, N. K., Caris, E., Conwell, J., Edwards, L. A., Lewin, M., Pinto, N. M., Studer, M., Young, L., & Arya, B. (2026). Socioeconomic and geographic factors are associated with lower prenatal diagnosis rates of

- congenital heart disease. *Progress in Pediatric Cardiology*, 101948, [Impact Factor 0.4], Cite, Source Document)
4. Willson, R., Thomsen, S. R., Swanson, M., **Howard, W. J.**, Mistovich, R. J., Ellis, H. B., ... & Schmale, G. A. (2026). No Difference in Posterior Tibial Slope Comparing Patients With and Without Delayed Anterior Cruciate Ligament Tear After Tibial Spine Fracture: A Multicenter Study. *Journal of the Pediatric Orthopaedic Society of North America*, 15, 100314. [Impact Factor 1.1] (PDF, Cite, Source Document)
 5. Ohlsen, S., Son, M., **Howard, W. J.**, Sheasley, J., Niu, E., Kirby, J., Gopalan, M., Stavinoha, T., Carsen, S., Stinson, Z., Finlayson, C., Nault, M. L., Lee, R. J., Haus, B., Green, D., Schlechter, J., Heyworth, B., Beck, J., Nguyen, J., PRiSM Meniscus Research Interest Group, & Schmale, G. (2026). Risk Factors for Complication and Reoperation in the Treatment of Discoid Lateral Meniscus in Children and Adolescents: A Multicenter Study. *Orthopaedic Journal of Sports Medicine*, [Impact Factor 2.5], Cite, Source Document)
 6. Kim, B., **Howard, W. J.**, Weiss, E. (2026). Diverging Illness Perceptions in the NICU: Insights from Parents, Physicians, and Nurses. *Journal of Perinatology* [Impact Factor 2.4]
 7. Nicholson, J., Deboeck, P. R., Gomer, B. & **Howard, W. J.** (2026). Implications of Missing at Random Assumption in Cognitive Training Randomized Controlled Trials: Current Practices and Suggestions. *Journal of the American Geriatrics Society* [Impact Factor 4.5], (PDF, Cite, Source Document)
 8. McKinney, C., **Howard, W. J.**, Bijlani, K., Kinter, S., Crerand, C., Dang, R., Jackson, O., Discolo, C., Rosenman, M., Mendonca, E., Dickinson, K., Heike C. (2026). Growth in Children with a Cleft Lip and/or Palate in 8 US Hospitals: A Retrospective Cohort Study. *JDR Clinical & Translational Research*, 21(6), 1007-1021. [Impact Factor 2.2], Cite, Source Document)
 9. Rea H. M., Lindley, A., **Howard, W. J.**, Neuhaus E., Faja, S., Bernier, R., Chawarska, K., Dawson, G., Dziura, J., Jeste, S., Kleinhans, N., Naples, A. J., Nelson, C. A., Sabatos-Devito, M., Seow, H., Shic, F., Sugar, C. A., McPartland, J. C., & Webb, S. J. (2026). A longitudinal investigation of aggression and social skills in youth with autism spectrum disorder. *Research in Autism*, 130, 202767. [Impact Factor 3.4], PDF, Cite, Source Document)
 10. Willson, R., Thomsen, S., Swanson, M., **Howard, W. J.**, Mistovich, J., Ellis, H., Cruz, A., Jr., Fabricant, P., Lee, J. R., McKay, S., Yen, Y.-M., Kushare, I. V., Ganley, T. J., & Schmale, G. A. (2025). No impact of posterior tibial slope on subsequent ACL tears in patients with tibial spine fractures, 100314. *Journal of the Pediatric Orthopaedic Society of North America*, 100314. Cite, Source Document)
 11. Zhai, S., Palermo, T. M., Shenoi, S., Demiris, G., **Howard, W. J.**, Kientz, J., Yuwen, W., Ward, T. M. (2025). A shared-management web-based intervention for sleep deficiency in school-age children with juvenile idiopathic arthritis and their parents: feasibility and acceptability study. *Journal of Clinical Sleep Medicine*, 21(6),

- 1007-1021. [Impact Factor 4.2], Cite, Source Document)
12. Dervan, L., Garros, D., **Howard, W. J.**, & Roumeliotis, N. (2025). Prevalence and Predictors of Home Discharge from the Pediatric Intensive Care Unit. *Critical Care Explorations*, 7(5), e1266. [Impact Factor 2.7] (PDF, Cite, Source Document)
 13. McKinney, C., **Howard, W. J.**, Bijlani, K., Kinter, S., Crerand, C., Dang, R., Jackson, O., Discolo, C., Rosenman, M., Mendonca, E., Dickinson, K., Heike C. (2025). A Retrospective Cohort Study of Growth in Children with Orofacial Cleft in 8 US Hospitals. *Journal of the Academy of Nutrition and Dietetics*, 125(4), 537-544. [Impact Factor 4.0] (PDF, Cite, Source Document)
 14. Ward, T., **Howard, W. J.**, & Palermo, T. (2025). A Shared-Management Web-Based Intervention for Sleep Deficiency in School-Age Children with Juvenile Idiopathic Arthritis and their Parents: Feasibility and Acceptability Study. *Journal of Clinical Sleep Medicine* [Impact Factor 4.2] (Source Document)
 15. Ross, F., Tjoeng, L., Chan, T., Tan, J., **Howard, W. J.**, & Jimenez, N. (2025). The Impact of Medicaid Expansion on Congenital Heart Surgery Outcomes: Data from the Society of Thoracic Surgeons Database. *Anesthesia & Analgesia* [Impact Factor 4.6] (Source Document)
 16. Kim, S., Kim, L., **Howard, W. J.**, McNulty, B., Salehi, P. (2024). Aberrant Behavior Checklist in Youth with Prader-Willi Syndrome: Preliminary Study of Cross-sectional and Longitudinal Behavior Characterization. *American Journal of Medical Genetics Part A* [Impact Factor 2.249] (Source Document)
 17. Webb, S. J., **Howard, W. J.**, Garrison, M., Corrigan, S., Quinata, S., Taylor, L., & Christakis, D. A. (2024). Mobile Media Content Exposure and Toddlers' Responses to Attention Prompts and Behavioral Requests. *JAMA Network Open*, 7(7), e2418492-e2418492. [Impact Factor 13.8] (PDF, Cite, Source Document)
 18. McKinney, C. M., **Howard, W. J.**, Bijlani, K., Rahman, M., Meehan, A., Evans, K. N., Leavitt, D., Sitzman, T. J., Amoako-Yirenkyi, P., & Heike, C. L. (2024). Growth patterns between 0 to 36 months of age among US children with orofacial cleft: A retrospective cohort study. *Journal of the Academy of Nutrition and Dietetics*. [Impact Factor 4.7] (PDF, Cite, Source Document)
 19. Law, E. F., Tham, S. W., **Howard, W. J.**, Ward, T. M., & Palermo, T. M. (2024). Executive Functioning and Self-Management Processes Mediate the Relationship between Insomnia and Pain-Related Disability. *The Journal of Pain*, 25(1), 273-283. [Impact Factor 5.559] (PDF, Cite, Source Document)
 20. Boyd, B. A., **Howard, W. J.**, Bodfish, J. W., Lecavalier, L., Harrop, C., Jones, D., Dallman, A., Kalburgi, S. N., & Hollway J. (2024). Examining Sensitivity to Developmental Changes on the Behavioral Inflexibility Scale. *Journal of Autism and Developmental Disorders*, 1-10. [Impact Factor 5.258] (PDF, Cite, Source Document)
 21. Okech, D., **Howard, W. J.**, Matthew, R. A., & Purser, G. L. (2020). The effects of sociodemographic variables on the economic behaviour of poorer households

- in the US and Kenya: a cross-national analyses. *Journal of Contemporary African Studies*, 1-19. [Impact Factor 0.69] (PDF, Cite, Source Document)
22. Okech, D., Hansen, N., **Howard, W. J.**, Anarfi, J. K., & Burns, A. C. (2018). Social support, dysfunctional coping, and community reintegration as predictors of PTSD among human trafficking survivors. *Behavioral Medicine*, 44(3), 209-218. [Impact Factor 2.442] (PDF, Cite, Source Document)
 23. Desai, A. D., Jacob-Files, E. A., Lowry, S. J., Opel, D. J., Mangione-Smith, R., Britto, M. T., & **Howard, W. J.** (2018). Development of a Caregiver-Reported Experience Measure for Pediatric Hospital-to-Home Transitions. *Health Services Research. Behavioral Health Services Research*, 53(S1), 3084-3106. [Impact Factor 1.769] (PDF, Cite, Source Document)
 24. Malik, F. S., Yi-Frazier, J. P., Taplin, C. E., Roth, C. L., Whitlock, K. B., **Howard, W. J.**, & Pihoker, C. (2018). Improving the Care of Youth with Type 1 Diabetes with a Novel Medical-Legal Community Intervention: The Diabetes Community Care Ambassador Program. *The Diabetes Educator*, 44(2) 168-177. [Impact Factor 1.736] (PDF, Cite, Source Document)
 25. Okech, D., McGarity, S. V., Hansen, N., Burns, A. C., & **Howard, W. J.** (2018). Financial capability and sociodemographic factors among survivors of human trafficking. *Journal of Evidence-Informed Social Work*, 15(2), 123-136. (PDF, Cite, Source Document)
 26. Stein, E., **Howard, W. J.**, Rowhani-Rahbar, A., Rivara, F. P., Zatzick, D., & McCarty, C. A. (2017). Longitudinal trajectories of post-concussive and depressive symptoms in adolescents with prolonged recovery from concussion. *Brain injury*, 31(13-14), 1736-1744. [Impact Factor 2.061] (PDF, Cite, Source Document)
 27. Simon, T. D., Whitlock K. B., Haaland, W., Wright, D. R., Hood, M., Zhou, C., Neff, J., **Howard, W. J.**, Cartin, B., & Mangione-Smith, R. (2017). Effectiveness of a comprehensive case management service for children with medical complexity. *Pediatrics*, 140(6) e20171641. [Impact Factor 5.515] (PDF, Cite, Source Document)
 28. Law, E. F., Fisher, E., **Howard, W. J.**, Levy, R., Ritterband, L., & Palermo, T. M. (2017). Longitudinal change in parent and child functioning after internet-delivered cognitive-behavioral therapy for chronic pain. *PAIN*, 158(10), 1992-2000. [Impact Factor 5.559] (PDF, Cite, Source Document)
 29. Nicholson, J., Deboeck, P. R., & **Howard, W. J.** (2017). Attrition in developmental psychology: A review of modern missing data reporting and practices. *International Journal of Behavioral Development*, 41(1) 143-153. [Impact Factor 1.760] (PDF, Cite, Source Document)
 30. **Howard, W. J.**, Rhemtulla, M., & Little, T. D. (2015). Using principal component analysis (PCA) to obtain auxiliary variables for missing data in large data sets. *Multivariate Behavioral Research*, 50(3), 285-299. [Impact Factor 3.691] (PDF, Cite, Source Document)

31. Chan, C. S., Rhodes, J. E., **Howard, W. J.**, Lowe, S. R., Schwartz, S. E., & Herrera, C. (2013). Pathways of influence in school-based mentoring: The mediating role of parent and teacher relationships. *Journal of School Psychology, 51*(1), 129-142. [Impact Factor 2.299] (PDF, Cite, Source Document)
32. Greenwood, C. R., Buzhardt, J., Walker, D., McCune, L., & **Howard, W. J.** (2013). Advancing the construct validity of the Early Communication Indicator (ECI) for infants and toddlers: Equivalence of growth trajectories across two early head start samples. *Early Childhood Research Quarterly, 28*(4), 743-758. [Impact Factor 2.364] (PDF, Cite, Source Document)
33. Greenwood, C. R., Walker, D., Buzhardt, J., **Howard, W. J.**, McCune, L., & Anderson, R. A., (2013). Evidence of a continuum in foundational expressive communication skills. *Early Childhood Research Quarterly, 28*, 540-554. [Impact Factor 2.364] (PDF, Cite, Source Document)
34. Okech, D., **Howard, W. J.**, Mauldin, T., Mimura, Y., & Kim, J. (2012). The Effects of Economic Pressure on the Resilience and Strengths of Individuals Living in Extreme Poverty. *Journal of Poverty, 16*(4), 429-446. (PDF, Cite, Source Document)
35. Okech, D., **Howard, W. J.**, & Kim, J. (2012). Efficacy and agency among poor families with and without children. *Child & Family Social Work, 17*(2) 1-12. (PDF, Cite, Source Document)
36. Buzhardt, J., Greenwood, C. R., Walker, D., Anderson, R., **Howard, W. J.**, & Carta, J. J. (2011). Effects of web-based support on early head start home visitors use of evidence-based intervention decision making and growth in children's expressive communication. *NHSA Dialog: A Research-to-Practice Journal for the Early Childhood Field, 14*(3), 121-146. (PDF, Cite, Source Document)
37. Greenwood, C. R., Buzhardt, J., Walker, D., **Howard, W. J.**, & Anderson, R. A. (2011). Program-level influences on the measurement of early communication of infants and toddlers in Early Head Start. *Journal of Early Intervention, 33*(2), 110-134. [Impact Factor 1.241] (PDF, Cite, Source Document)

Software

Howard, W. J. (2023, November). An automated tool for creating NIH/IES/NSF/PCORI biosketches [Computer software]. Available from www.waylon.github.io.

Howard, W. J. (2022, May). An intuitive tool for plotting power curves to assess model fit using RMSEA [Computer software]. Available from www.waylon.github.io

Howard, W. J. (2022, March). An intuitive interactive tool that enhances team collaboration by streamlining task tracking and management [Computer software]. Available from www.waylon.github.io.

Howard, W. J. (2021, April). An intuitive interactive tool that helps users quickly generate reliable project estimates for better planning and decision-making [Computer software]. Available from www.waylon.github.io.

Howard, W. J. (2020, April). An interactive tool for comparing nested models using the Satorra-Bentler chi-square test. [Computer software]. Available from www.waylon.github.io.

Howard, W. J. (2020, April). An interactive tool for comparing nested models using the Satorra-Bentler chi-square test. [Computer software]. Available from www.waylon.github.io.

Presentations

Oral Presentations and Panel Discussions

Wissow, L., King, S., Cunningham, K. A., Warner, E., Bijlani, K., & **Howard, W. J.** (2026, October). Drivers of Change in Who Is Served by Pediatric Integrated Care. In AACAP's 2026 Annual Meeting. AACAP. ([link](#))

Paek, B., **Howard, W. J.**, Keller, S., Hirose, S., Dorn, M., & Walker, M. (2025). TTTS: Likelihood of donor survival—Introducing a calculator for prediction of donor demise on post-op day 1 and donor live birth. International Fetal Medicine & Surgery Society (IFMSS) Annual Meeting, Napa, CA, USA.

Howard W. J. (2025, June) Streamlined Workflows for Statistician–Investigator Research Partnerships [In-Person Invited Talk]. Center for Child Health Behavior and Development, Seattle Childrens Research Institute, Seattle, WA.

Willson, R., Thomsen, S., Swanson, M., **Howard W. J.**, Mistovich, J., Ellis, H.B., Cruz Jr., A., Fabricant, P., Lee, R.J., McKay, S., Yen, Y-M., Kushare, I.V., Ganley, T.J., Schmale, G.A. (2025, January). No Impact of Posterior Tibial Slope on Subsequent ACL tears in Patients with Tibial Spine Fractures [In-Person Contributed Talk]. 12th Annual Meeting of the Pediatric Research in Sports Medicine Society, Columbus, OH. ([link](#))

Howard W. J. (2024, June) SEM, Revealed. Modeling connections with latent variables and regression pathways [In-Person Invited Talk]. Center for Child Health Behavior and Development, Seattle Childrens Research Institute, Seattle, WA.

McKinney, C., **Howard W. J.**, Bijlani, K., Evans, K., Kassuhn, M., Leavitt, D., Heike, C. (2023, May). Growth velocity in U.S. children with orofacial clefts in the first 6 months of life [In-Person Contributed Talk]. 80th Annual Meeting of the American Cleft Palate Craniofacial Association (ACPA), Raleigh, NC. ([link](#))

McNulty, B., Salehi, P., Kim, L. Kim, S., **Howard W. J.** (2023, September). Aberrant Behavior Checklist Scores in Youth with Prader-Willi Syndrome [In-Person Contributed Talk]. Annual Foundation for Prader-Willi Research Symposium and Family Conference, Atlanta, GA. ([link](#))

Okech D., **Howard, W. J.**, (2018, January). Predictors of Financial Capability among Female Survivors of Human Trafficking in Ghana, West Africa [In-Person Symposia]. Financial Capability across the Globe: Results from Canada, China, and Ghana, 22nd

Annual Conference of the Society for Social Work and Research, Washington D.C. (link)

Howard, W. J. (2017, April). Missing Data in Applied Research [In-Person Invited Talk]. University of Washington, Dept. of Epidemiology, Seattle, WA. (outline)

Howard, W. J. (2015, October). Planned Missing Data in Pediatric Research [In-Person Invited Talk]. Seattle Childrens Research Institute, Center for Child Health Behavior and Development, Seattle, WA. (outline)

Howard, W. J., Johnston, H., & Misra, C., (2015, May). Research Computing for Social Scientists [In-Person Roundtable Discussions]. University of Massachusetts, Amherst, MA. (outline, link)

Howard, W. J. (2015, February). Planned Missing Data Designs [In-Person Invited Talk]. Department of Statistics, University of Massachusetts, Amherst, MA.

Howard, W. J. (2014, March). Missing Data [In-Person Invited Talk]. Institute for Social Science Research, University of Massachusetts, Amherst, MA.

Howard, W. J. (2014, February). Introduction to Biostatistics [In-Person Symposia]. Lecture for the James A. Ferguson Emerging Infectious Diseases Research Initiatives for Student Enhancement (RISE) Fellowship Program at the Kennedy Krieger Institute, Baltimore, MD. (link)

Howard, W. J. (2013, August). An alternative to the restrictive strategy in modern missing data procedures [In-Person Invited Talk]. Department of Mental Health, Johns Hopkins School of Public Health, Baltimore, MD.

Howard, W. J.. (2013, June). Using principal components analysis to obtain auxiliary variables for missing data in large data sets [In-Person Invited Talk]. Department of Psychology, University of North Carolina, Chapel Hill, NC.

Howard, W. J. & Little, T. D. (2013, May). Use of principal components analysis to reap the benefits of the inclusive strategy in the presence of many possible auxiliary variables [In-Person Invited Talk]. Annual Modern Modeling Methods (M3) Conference, Storrs, CT.

Howard, W. J. (2013, March). Missing data analysis [In-Person Invited Talk]. Bi-annual Biostatistics, Epidemiology and Data Management Core, Johns Hopkins University, Bayview, MD.

Howard, W. J. (2012, November). Understanding missing data: An introduction to basic concepts [In-Person Invited Talk]. Bi-annual Biostatistics, Epidemiology and Data Management Core Seminar, Johns Hopkins University, Bayview, MD.

Howard, W. J. (2011, March). Latent growth modeling of language and early literacy for two parallel processes [In-Person Special Session]. Annual meeting of the Midwestern Social Development Consortium, Lawrence, KS.

Howard, W. J. & Anderson, R. A. (2010, April). Receiver operating characteristic curve benchmark decision-making [In-Person Flash Talk]. Quantitative Training Program Weekly Colloquium Series, University of Kansas, Lawrence, KS.

Howard, W. J., Ratliff, J. M., Turner, A. D., Smith, M., & Page, M. C. (2008, May). Summer research programs: Getting in and making the most of your experience [In-Person Roundtable Discussion]. 20th annual meeting of the Association for Psychological Science, Chicago, IL.

Evans, S. J., Baker, A. J., **Howard, W. J.**, & Roberts, S. L. (2008, April). Exploring graduate school: Preparation, application, and experiences [In-Person Roundtable Discussion]. 54th annual meeting of the Southwestern Psychological Association, Kansas City, MO.

Long, R. L., **Howard, W. J.** & McIntosh, A. (2008, November). Internationalize your vitae: How to have a global career in psychology [In-Person Roundtable Discussion]. Joint meeting of the Association for Psychological and Educational Research in Kansas and the Nebraska Psychological Society, Bellevue, NE.

Interactive Sessions

Yeji, B. & **Howard W. J.** (2026, February) Linear Mixed Models (LMMs) and Applied R Packages [Webinar Workshop]. Research Forum Training Session. Statistical Literacy for Biologists, Seattle Childrens Research Institute, Seattle, WA.

Howard W. J. (2025, March) Latent variable modeling in behavioral and clinical outcomes research: A practical guide [Webinar Workshop]. Research Forum Training Session. Behavioral and Clinical Outcomes Research Program, Seattle Childrens Research Institute, Seattle, WA.

Howard W. J. (2023/2024, November) An introduction to missing data analysis [Webinar Workshop]. Statistical Literacy for Biologists Training Session. Research Scientific Computing, Seattle Childrens Research Institute, Seattle, WA.

Howard, W. J. (2015, June) Introduction to Confirmatory Factor Analysis [In-Person Workshop]. Summer Summit Research Methodology Conference, University of Massachusetts, Institute for Social Science Research, Amherst, MA. (outline, link)

Howard, W. J. (2011, February). Multilevel modeling in MULTILEV [In-Person Tutorial]. Saturday Seminar, Center for Research Methods and Data Analysis, University of Kansas, Lawrence, KS.

Howard, W. J., & McCune, L. A. (2011, October). Latent variable modeling using LISREL notation [In-Person Tutorial]. Saturday Seminar, Center for Research Methods and Data Analysis, University of Kansas, Lawrence, KS.

Howard, W. J. (2010, September). Latent variable modeling using LISREL notation [In-Person Tutorial]. Saturday Seminar, Center for Research Methods and Data Analysis, University of Kansas, Lawrence, KS.

Visual Presentations

Ormanidou, I., Howard, W., Bijlani, K., Denno, D. M., Amoako-Yirenkyi, P., Heike, C., Owusu Adusei, A., Oliphant, H., Crerand, C. E., Jackson, O., Discolo, C., Vetter, S., Wilson, K., Geronimo, Y., Galera, R., Lyman, N., Whyte, V., Pajor, N., & McKinney, C. M. (2027). Identifying factors that predict underweight in infants with cleft lip and/or palate in 10 US children's hospitals [Conference abstract]. American Cleft Palate-Craniofacial Association Annual Meeting.

Ormanidou, I., Howard, W., Bijlani, K., Denno, D., Amoako-Yirenkyi, P., Kinter, S., Heike, C., & McKinney, C. (2026, November 5–7). Development and validation of a multivariable prediction model for underweight status at 4 months in infants with cleft lip and/or palate across U.S. children's hospitals [Accepted poster presentation]. 11th MIOHR Congress, Athens, Greece.

Roome, K., Arya, B., Opel, D., & **Howard, W. J.** (2026, April). Current practices and views: Prenatal critical congenital heart disease diagnoses & palliative care involvement. *Journal of the American College of Cardiology*, 87(13 Suppl.), A375. (link)

Schauer, J., Zheyi, T., Wagner, T., Spencer, K., Rivers, J., Chisholm, K., Smith, J., Saarela, K., Albers, E., Hong, B., Kemna, M., Kronmal, R., **Howard, W. J.**, & Law, Y. (2026, March). Monitoring for post-transplant lymphoproliferative disease in pediatric patients by quantifying Epstein Barr Virus viral load. 46th Annual Meeting & Scientific Sessions of the International Society for Heart and Lung Transplantation (ISHLT), Toronto, Canada.

Roome, K., Arya, B., Opel, D., & **Howard, W. J.** (2026, March). Current practices and views: Prenatal critical congenital heart disease diagnoses & palliative care involvement. 46th Annual Meeting & Scientific Sessions of the International Society for Heart and Lung Transplantation (ISHLT), Toronto, Canada.

Ormanidou, I., **Howard, W. J.**, Bijlani, K., Denno, D., Amoako-Yirenkyi, P., Kinter, S., & Heike, C., & McKinney, C. (2026, March 25–28). Predicting underweight in children with cleft across 10 hospitals. 26th International Association for Dental Research (IADR) General Session & Exhibition, San Diego, CA, USA.

McKinney, C. M., Kinter, S., Ormanidou, I., **Howard, W. J.**, Bijlani, K., Madhoun, L., Crerand, C., Meehan, A., Rahman, M., & Heike, C. L. (2025, October 20–24). To what extent do children with a cleft follow World Health Organization growth standards in the first 2 years of life? 15th International Congress on Cleft Lip/Palate and Related Craniofacial Anomalies, Kyoto, Japan.

McKinney, C., **Howard, W. J.**, Bijlani, K., Rahman, A., Meehan, K., Evans, D., Leavitt, T. J., Sitzman, P., Amoako-Yirenkyi, C. L., Heike, C. L. (2025, May). Impact of Comorbidities on Growth in Children With Cleft From 0-18 Months of Age. Annual Meeting of the American Cleft Palate Craniofacial Association (ACPA), Palm Springs, CA. (link)

Buzhardt, J., Nowell, S., Higgins, S., & **Howard, W. J.** (2025, October 7). A novel play-based measure of social-communication for autistic children [Abstract]. Divi-

sion for Early Childhood Annual International Conference on Young Children with Disabilities and Their Families, Portland, OR, USA.

Lyle, A., **Howard, W. J.**, Weiss, E. (2025, April). Differences in Comfort in Ethical Dilemmas Among Pediatric Residents: A Survey Study. Poster presented at the annual Pediatric Academic Societies (PAS) Conference, Honolulu, HI. ([link](#))

Ittleman, B., **Howard, W. J.**, Virk, K., Brown, N., Bhat, A., Caris, E., Conwell, J., Edwards, L., Lewin, M., Pinto, N., Studer, M., Young, L., & Bhawna, A. (2024, November). Socioeconomic and Geographic Factors Associated with Reduced Prenatal Diagnosis of Significant Congenital Heart Disease at a Single Regional Institution. Poster presented at the centennial American Heart Association Scientific Sessions, Chicago, IL. ([link](#))

Lyle, A., **Howard, W. J.**, Weiss, E. (2024, May). Pediatric Residents Perceptions of Ethics Education in the Neonatal Intensive Care Unit: A Needs Assessment. Poster presented at the annual Pediatric Academic Societies (PAS) Conference, Toronto, ON, CA. ([link](#))

McKinney, C., **Howard W. J.**, Bijlani, K., Rahman, A., Meehan, K. Evans, D. Leavitt, T. J. Sitzman, P. Amoako-Yirenkyi, C. L. Heike (2024, May). Growth in U.S. Infants with Orofacial Cleft. 31st Annual Meeting of the American Society of Craniofacial Surgeons, Denver, CO. ([link](#))

Chrisman, S., Tham, S., **Howard W. J.**, Zatzick, D. F., Rivara, F. P., McCarty, C. M. (2023). Sleep and Concussion Recovery in a Pediatric Collaborative Care Intervention Trial: Do Concussion Symptoms Improve Because Teens are Sleeping Better? Poster presented at the annual Neurotrauma conference, Austin, TX. ([link](#))

McNulty, B., Kim, L. Kim, S., **Howard W. J.**, Salehi, P. (2023). Changes in hyperphagia behaviors correlate with both improvement in quality of life and worsening of maladaptive behaviors in children with Prader-Willi Syndrome over time. Poster presented at the annual Pediatric Academic Societies (PAS), Washington, D.C. ([link](#))

Zhou, C., Gao, Y., **Howard, W. J.** (2017). Evaluation of combining bootstrap with multiple imputation using R on Knights Landing platform. Poster presented at the 4th International Conference on Cyber Security and Cloud Computing, New York, New York. (PDF, [link](#))

Malik, F. S., Yi-Frazier, J. P., Taplin, C. E., Roth, C. L., Whitlock, K., **Howard, W. J.**, Pihoker, C. (2017). Psychosocial Mediators Impacting the Effectiveness of a Novel Medical-Legal Community Care Program. Poster presented at the Annual Meeting of the American Diabetes Association, San Diego, California. ([link](#))

Desai, A. D., Opel, D. J., Mangione-Smith R., Britto, M. T., & **Howard, W. J.** (2015). Development of a Caregiver-Reported Outcome Measure for Pediatric Hospital-to-Home Transitions. Poster presented at the Annual Pediatric Academic Societies Meeting, San Diego, California.

Matson, P., **Howard, W. J.**, Chung, S. (2014). Use of P-technique to examine the dynamic relationship among STD-associated feelings and perceptions in a cohort of adolescent females. Poster presented at the 47th Annual Meeting of the Society for Epidemiological Research, Seattle, Washington. (link)

Stone, J., **Howard, W. J.**, & DAbundo M., L. (2013). A latent variable approach to attitudes and practices of health professionals serving persons with developmental disabilities. Poster presented at the 42nd annual meeting of the Association of University Centers on Disabilities, Washington, DC.

Greenwood, C. R., Carta, J. J., Atwater, J., **Howard, W. J.**, & McCune L. A. (2012). Program and ELL differences in the quantity and quality of preschool language and early literacy instruction and child Outcomes. Poster presented at Head Starts 11th National Research Conference, Washington, DC. (link)

Chan, C. S., Lowe, S. R., **Howard, W. J.**, Schwartz, S. E. O., & Rhodes, J. E. (2012). Pathways of influence in school-based mentoring: The mediating role of parent and teacher relationships. Poster presented at the 14th biennial meeting of the Society for Research on Adolescence (SRA), Vancouver, British Columbia, Canada. (link)

McCune A. L., **Howard, W. J.**, Greenwood, C. R., Walker, D., Buzhardt, J., & Anderson, R. A. (2012). Dynamic relations within and between early communication key skill elements over time: Evidence of contiguous continuity in skill development. Poster presented at the Developmental Methodology Themed Meeting of the Society for Research in Child Development, Tampa, FL. (link)

Buzhardt, J., Greenwood, C. R., Walker, D., Anderson, R. A., & **Howard, W. J.** (2011). A web based tool to support data-based intervention decision-making for infant and toddler language growth. Poster presented at the annual meeting of the Kansas Association for Behavior Analysis, Overland Park, KS.

Okech, D., & **Howard, W. J.** (2011). Down but not out: Resilience in poor households. Poster presented at the 15th annual meeting of the Society for Social Work and Research, Tampa, FL. (link)

Carta, J. J., Greenwood, C. R., Pitchlyn, C., & **Howard, W. J.** (2010). Challenging behavior implications for childrens early communication and literacy development. Poster presented at the 26th annual meeting of the Division for Early Childhood of the Council for Exceptional Children, Kansas City, KS.

Geldhof, G. J., Stump, K., **Howard, W. J.**, & Hawley P. H. (2010). Support for five distinct resource control types via a cluster analytic approach. Poster presented at the 12th biennial meeting of the Society for Research on Adolescence, Philadelphia, PA. (link)

Greenwood, C. R., Buzhardt, J., Walker, D., Anderson, R. A., & **Howard, W. J.** (2010). The early communication indicator (ECI) for infants and toddlers: Early head start growth norms. Poster presented at the 26th annual meeting of the Division for Early Childhood of the Council for Exceptional Children, Kansas City, KS.

Greenwood, C. R., Anderson, R. A., Walker, D., Buzhardt, J., & **Howard, W. J.** (2010). Exploring the early communication key skills for intervention decision-making with infants and toddlers. Poster presented at 5th annual Institute of Education Sciences Research Conference, National Harbor, MD.

Grant and Contract Activity

Selected (external)

Key Personnel (2024 - 2027): Developing a multi-dimensional measure of adolescent engagement in research. (PCORI), Contact PI/ Project Leader: Kroshus, Emily; Seattle Children's Research Institute (total project funding: \$710,000; FTE allocation: 10% for 2 years), Status: *Active*

Key Personnel (2025 - 2030): Evaluating Huddles as a Novel Approach to Improving Concussion Safety. (NIH), Contact PI/ Project Leader: Kroshus, Emily; Chrisman, Sara; Seattle Children's Research Institute (total project funding: \$4,344,318; FTE allocation: 10% for 5 years), Status: *Active*

Key Personnel (2025 - 2029): Evaluating and strengthening Special Olympics approach to sexual violence prevention. (CDC), Contact PI/ Project Leader: Kroshus, Emily; Seattle Children's Research Institute (total project funding: \$1,125,000; FTE allocation: 10% for 4 years), Status: *Active*

Key Personnel (2025 - 2030): Randomized trial of a shared decision-making intervention for families about firearm storage. (NIH Ro1), Contact PI/ Project Leader: Kroshus, Emily; Seattle Children's Research Institute (total project funding: \$4,390,219; FTE allocation: 10% for 5 years), Status: *Active*

Key Personnel (2025 - 2030): Sleep Health and Youth's Sociocultural and Neighborhood Context in Childhood Systemic Lupus Erythematosus. (NIH Ro1), Contact PI/ Project Leader: Jimenez, Nathalia; Seattle Children's Research Institute (total project funding: \$1,736,236; FTE allocation: 10% for 5 years), Status: *Active*

Key Personnel (2025 - 2030): Understanding Longitudinal Relationships between Sleep and Menstrual Pain in Adolescents. (NIH Ro1), Contact PI/ Project Leader: Tham, See Wan; Seattle Children's Research Institute (total project funding: \$4,416,992; FTE allocation: 10% for 5 years), Status: *Active*